

Team Name

LEGAL VISIONARIES

Project Title

LegalEase – AI-Powered Legal Document Generator

Team Members

1. **Rithika**
2. **Sindhuja**
3. **Sumathi**
4. **Sathvika**

PHASE 3: PROJECT DESIGN

1. Introduction

The Project Design phase defines the overall structure and working flow of the LegalEase system.

LegalEase is designed as a web-based AI-assisted legal document generation platform. The system accepts information from the user, processes the information through a backend API and Generative AI component, and produces a structured legal document.

The design uses a Streamlit frontend, FastAPI backend, Gemini AI integration, and document formatting components.

2. System Architecture

The proposed LegalEase architecture consists of the following major components: User



Streamlit Frontend



FastAPI Backend



Gemini AI



Generated Legal Document



Editable Preview



TXT / DOCX / PDF

Component Description Streamlit Frontend

Provides the user interface through which users enter document requirements and interact with the generated document.

FastAPI Backend

Receives user requests from the frontend and manages communication with the AI generation component.

Gemini AI

Processes the user's document requirements and generates the legal document content.

Document Formatting Components

Format the generated content for preview and export into TXT, DOCX, and PDF formats.

3. System Workflow

The LegalEase workflow is designed as follows:

Step 1 – User Input

The user selects a document type and enters relevant information such as:

- Parties involved
- Terms and conditions
- Effective date

Step 2 – Request Processing

The Streamlit frontend sends the user's information to the FastAPI backend.

Step 3 – AI Generation

The backend sends the required information to the Gemini AI generation component.

Step 4 – Document Creation

The AI-generated response is processed into a structured legal document.

Step 5 – Preview

The generated document is displayed to the user through an editable preview.

Step 6 – Editing

The user can review and modify the generated content.

Step 7 – Export

The final document can be exported into:

- TXT
- DOCX
- PDF

4. Module Design

The system is divided into the following major modules.

4.1 Frontend Module

Responsibility: User interaction and document preview.

Main functions:

- Document type selection
- Party information input
- Terms and conditions input
- Effective date input
- Generate Document button
- Editable document preview
- Download options

Responsible Team Member: Rithika – Frontend / UI

4.2 Backend Module

Responsibility: Processing requests and connecting the frontend with the AI generation component.

Main functions:

- Receive user requests
- Validate input
- Process document-generation requests
- Communicate with the AI component
- Return generated content

Responsible Team Member: Sindhuja – Backend

4.3 AI Integration Module

Responsibility: Generate legal document content using Generative AI.

Main functions:

- Receive document requirements
- Construct the required AI request
- Send the request to Gemini AI
- Receive generated content
- Return the generated document content

Responsible Team Member: Sumathi – AI Integration

4.4 Document Formatting and Export Module

Responsibility: Format generated content and provide downloadable documents.

The module supports:

- TXT generation
- DOCX generation
- PDF generation
- Professional formatting
- Document preview formatting

4.5 Testing and Documentation Module

Responsibility: Verify system functionality and maintain project documentation.

Main activities:

- Test user inputs
- Test document generation
- Test editing functionality
- Test TXT/DOCX/PDF exports
- Record test results
- Maintain project documentation

Responsible Team Member: Sathvika – Testing & Documentation

5. Frontend Design

The frontend will be implemented using Streamlit.

The interface will provide input fields for:

- Document Type
- Parties
- Terms and Conditions
- Effective Date

The interface will also provide controls for:

- Generate Document
- Edit Document
- Download TXT

- Download DOCX
- Download PDF

The generated document will be displayed in a structured preview area.

6. Backend Design

The backend will be implemented using FastAPI.

The backend will expose a document-generation endpoint.

Main Endpoint POST /generate

The endpoint will receive document-generation information such as:

Document Type

Parties

Terms

Effective Date

The backend will process the request and communicate with the AI generation component.

7. AI Integration Design

The AI integration component will connect the application with Gemini AI.

The basic process is:

User Requirements



FastAPI Backend



AI Generation Module



Gemini AI



Generated Legal Content



FastAPI Backend



Streamlit Frontend

The AI component will generate structured document content based on the information provided by the user.

8. Document Generation and Export Design

After AI generation, the document content will be processed for presentation and export.

TXT

Provides a simple text version of the generated document.

DOCX

Provides an editable Microsoft Word document.

PDF

Provides a formatted document suitable for viewing and sharing.

The project design also allows professional formatting elements such as headings, terms tables, logos, headers and footers.

9. Technology Stack Component Technology

Programming Language	Python
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Frontend	Streamlit
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Backend	FastAPI
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Server	Uvicorn
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AI	Gemini
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DOCX Generation	python-docx
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PDF Generation	FPDF
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Environment Configuration	python-dotenv
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Development Environment	Visual Studio Code / Python IDE
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Version Control	Git / GitHub
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10. Proposed Project Structure

The initial software structure is designed as follows:

LegalEase/

```
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|
|— main.py
|— routes.py
|— requirements.txt
|— .env
|
|— ai_core/
|   └─ gemini_generator.py
|
|   └─ formatting/
|       └─ docx_formatter.py
|       └─ pdf_formatter.py
|       └─ text_formatter.py
|
└─ frontend/
    └─ app.py
```

This structure separates the frontend, backend, AI integration, and documentformatting responsibilities.

11. Data Flow

The basic data flow of LegalEase is:

User



Input Details



Streamlit Frontend



FastAPI /generate



Gemini AI



Generated Document Content



Formatting



Editable Preview



Download

12. Security Design

The project will use environment variables for sensitive configuration such as the AI API key.

The API key should not be directly written inside the source code or uploaded publicly to GitHub.

User-provided information should be handled carefully during document generation.

13. Design Summary

The LegalEase system is designed as a modular web application consisting of a Streamlit frontend, FastAPI backend, Gemini AI integration, and document formatting/export components.

The design separates the responsibilities of user interaction, request processing, AI generation, document formatting, and testing.

This architecture provides a foundation for implementing the LegalEase application during the Project Development phase.